Huawei Access Node BAS-ACC-37 Standard Operating Procedure (SOP) for Routing Protocol Flap Recovery

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_9375

Description

This SOP outlines the procedures to diagnose and recover from a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure on the Huawei Access Node BAS-ACC-37 at the Basavanagudi POP. The routing protocol flap is a critical failure domain that can cause significant network instability and service disruptions. This SOP is designed to guide the network engineer through a series of diagnostic and resolution steps to isolate and fix the root cause of the problem. The underlying protocol mechanics of this failure domain are related to the routing protocol convergence and adjacency establishment processes. When a routing protocol flap occurs, the network device (BAS-ACC-37) may experience a high rate of routing updates, adjacency losses, and LSP flooding, leading to network instability and service disruptions. The Huawei Access Node is a critical component of the network infrastructure, and its failure can have a cascading impact on the entire network. Therefore, it is essential to follow this SOP to ensure prompt and effective recovery from a routing protocol flap failure.

Pre-requisite Checks

- Ensure that the Basavanagudi POP is accessible and that remote hands are dispatched to the site if necessary.

- Verify that the redundant link status is stable and functioning correctly.

- Check that the network device (BAS-ACC-37) is powered on and functioning properly.

Potential Root Causes

Cause	Probability
Incorrect BGP configuration	High
OSPF adjacency loss due to interface flap	Medium
ISIS LSP flooding due to misconfigured routing policies	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp process, Expected Output: BGP process
information, Description: This command displays the BGP process
information, including the BGP state, peer status, and routing table
information. This helps to identify any BGP-related issues and
troubleshoot the root cause of the problem.
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Command: display ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays the OSPF neighbor information, including the neighbor state, interface status, and LSA information. This helps to identify any OSPF-related issues and troubleshoot the root cause of the problem.

Command: display isis lsp, Expected Output: ISIS LSP information, Description: This command displays the ISIS LSP information, including the LSP state, LSP ID, and routing table information. This helps to identify any ISIS-related issues and troubleshoot the root cause of the problem.
```

Resolution Steps

Step 1: Step: Correct BGP configuration, Command: bgp 65535 neighbor 10.0.0.1 remote-as 65535, Description: This command corrects the BGP configuration by establishing a BGP session with the neighboring device. The remote-as option specifies the AS number of the neighboring device.

Step 2: Step: Re-establish OSPF adjacency, Command: ospf 10 area 0.0.0.0 interface GigabitEthernet 0/0/0 shutdown, Description: This command re-establishes the OSPF adjacency by shutting down the interface and then re-enabling it. This helps to re-establish the OSPF connection and resolve the adjacency loss issue.

Step 3: Step: Clear ISIS LSP flooding, Command: isis 10 lsp-clear, Description: This command clears the ISIS LSP flooding by resetting the LSP state and re-establishing the ISIS connection. This helps to resolve the ISIS LSP flooding issue and restore network stability.

Post-Check Verification

```
Command: display bgp process, Expected Output: BGP process information, Description: This command verifies that the BGP process is stable and functioning correctly. The expected output is the BGP process information, including the BGP state, peer status, and routing table information.

Command: display ospf neighbor, Expected Output: OSPF neighbor information, Description: This command verifies that the OSPF neighbor information is stable and functioning correctly. The expected output is the OSPF neighbor information, including the neighbor state, interface status, and LSA information.

Command: display isis lsp, Expected Output: ISIS LSP information, Description: This command verifies that the ISIS LSP information is stable and functioning correctly. The expected output is the ISIS LSP information, including the LSP state, LSP ID, and routing table information.
```

Roll-back Procedures

- Step: Restore original BGP configuration, Command: bgp 65535 neighbor 10.0.0.1 remote-as 65535 undo, Description: This command restores the original BGP configuration by undoing the BGP session establishment command. This helps to roll back the fix and return the network to its original state.

- Step: Revert OSPF adjacency establishment, Command: ospf 10 area 0.0.0.0 interface GigabitEthernet 0/0/0 undo shutdown, Description: This command reverts the OSPF adjacency establishment by undoing the interface shutdown command. This helps to roll back the fix and return the network to its original state.

- Step: Revert ISIS LSP clearing, Command: isis 10 lsp-clear undo, Description: This command reverts the ISIS LSP clearing by undoing the LSP clearing command. This helps to roll back the fix and return the network to its original state.